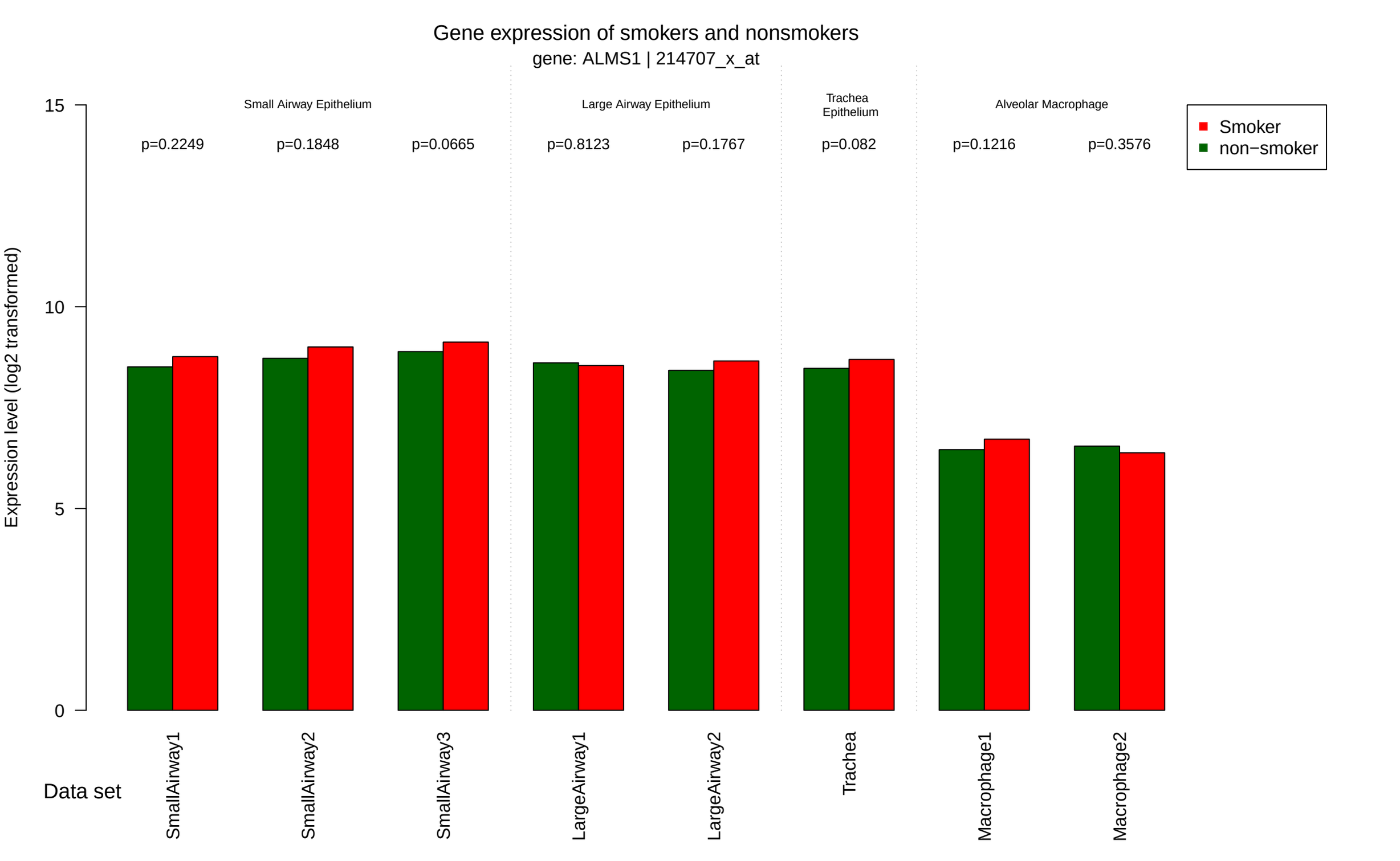
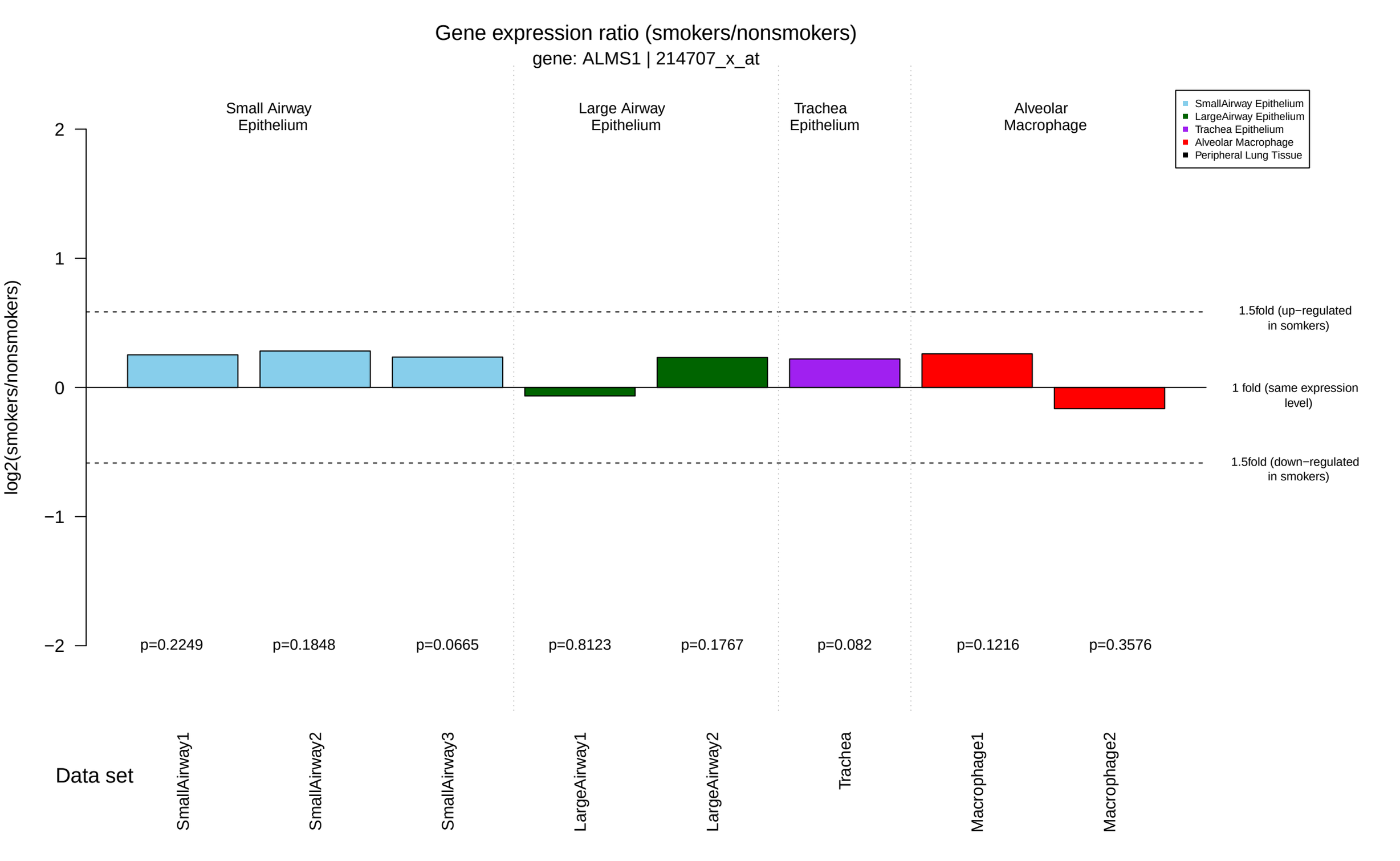
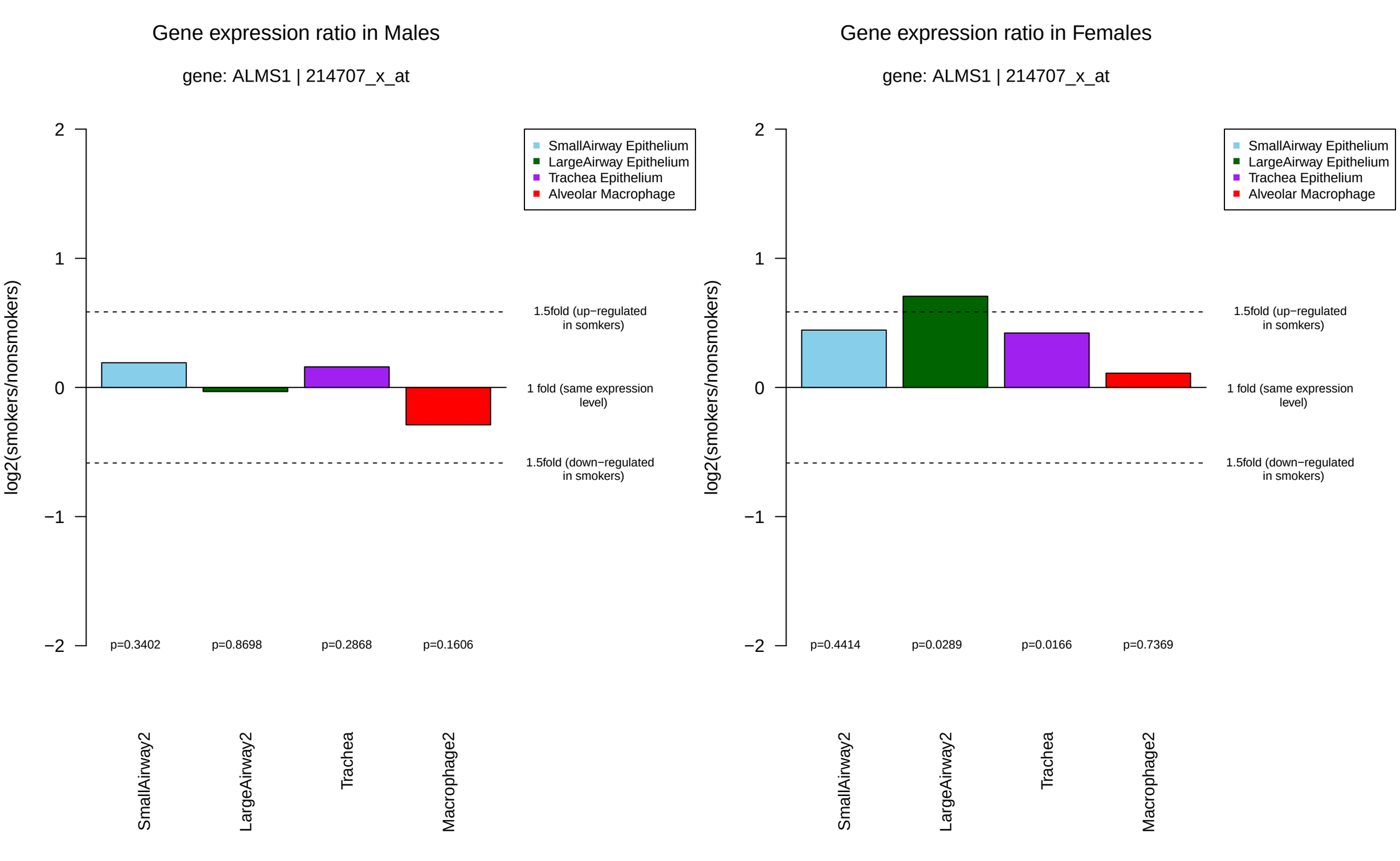
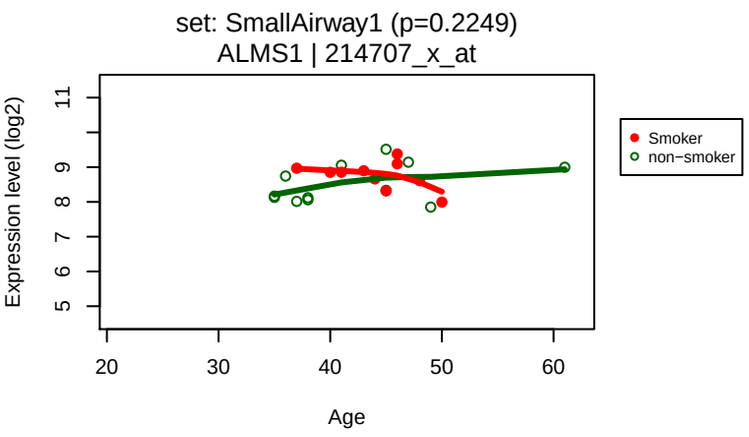
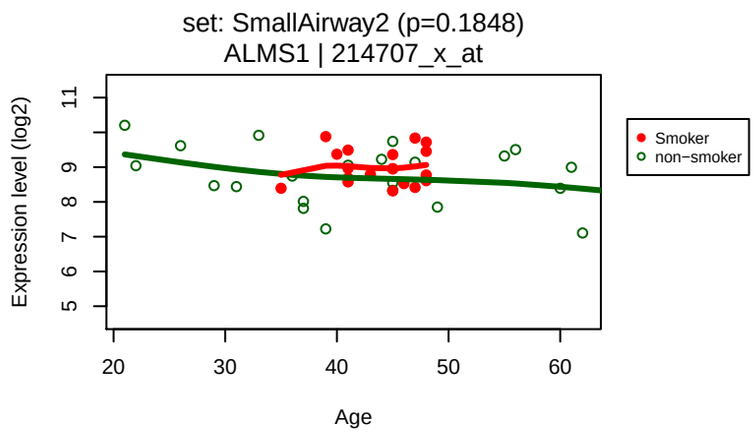
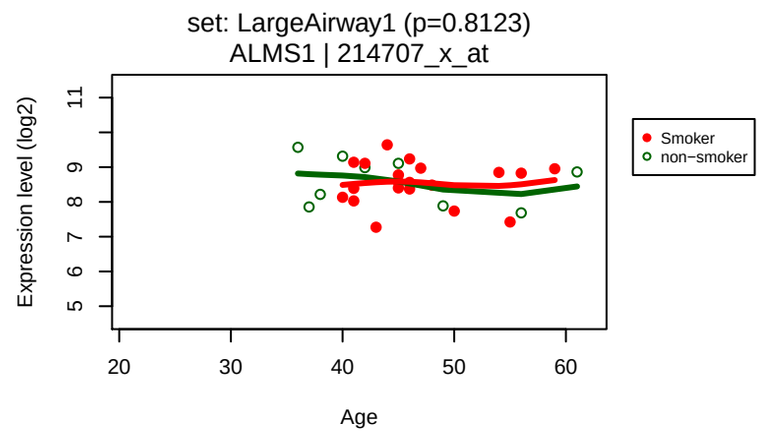

Gene expression of smokers and nonsmokers
gene: ALMS1 | 214707_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.2249
p=0.1848
p=0.0665
p=0.8123
p=0.1767
p=0.082
p=0.1216
p=0.3576
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

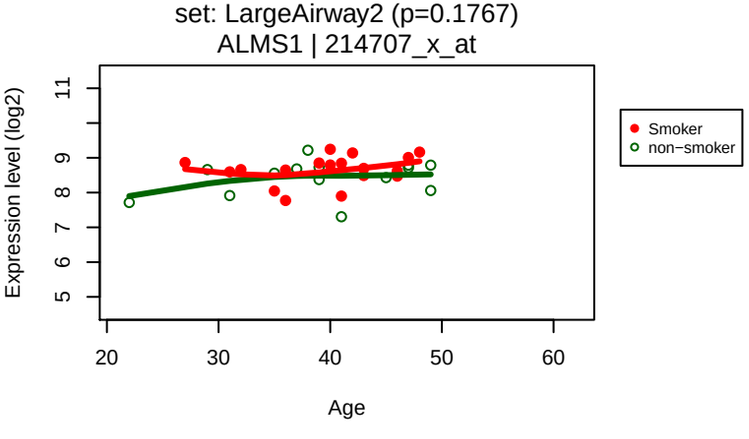

Gene expression ratio (smokers/nonsmokers)
gene: ALMS1 | 214707_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2249
p=0.1848
p=0.0665
p=0.8123
p=0.1767
p=0.082
p=0.1216
p=0.3576
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

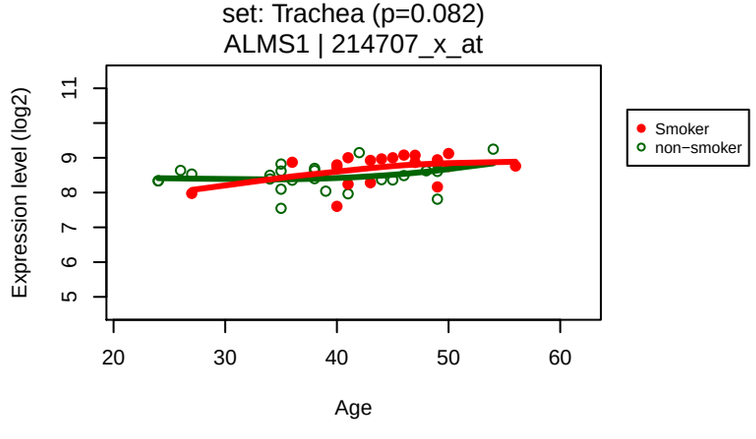

Gene expression ratio in Males
gene: ALMS1 | 214707_x_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.3402
p=0.8698
p=0.2868
p=0.1606
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: ALMS1 | 214707_x_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.4414
p=0.0289
p=0.0166
p=0.7369
SmallAirway2
LargeAirway2
Trachea
Macrophage2

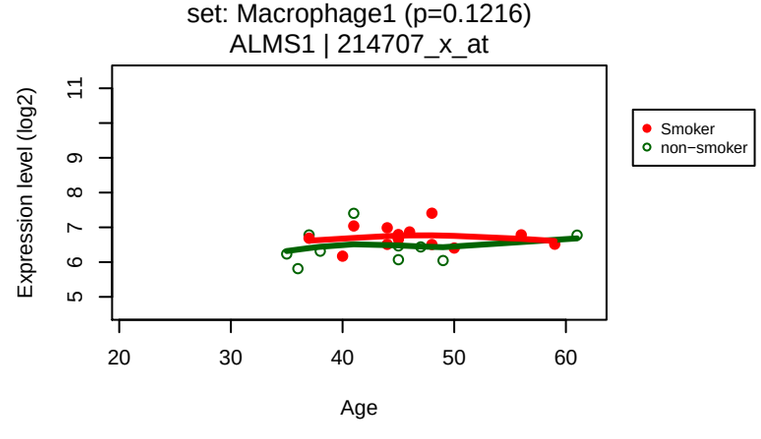

set: SmallAirway1 (p=0.2249)
ALMS1 | 214707_x_at
Expression level (log2)
Age
Smoker
non−smoker

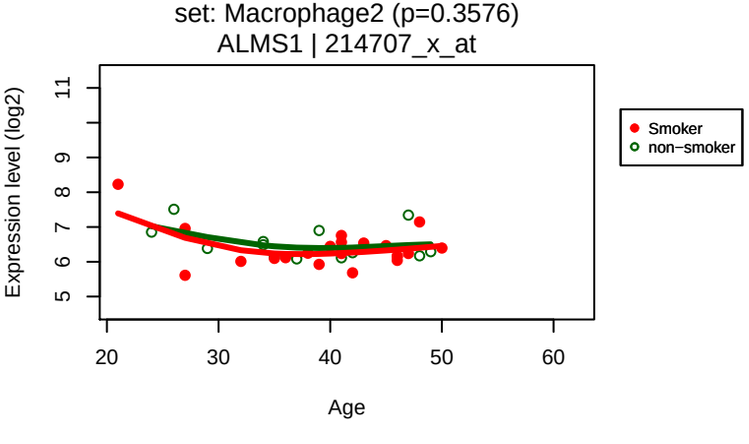

set: SmallAirway2 (p=0.1848)
ALMS1 | 214707_x_at
Expression level (log2)
Age
Smoker
non−smoker

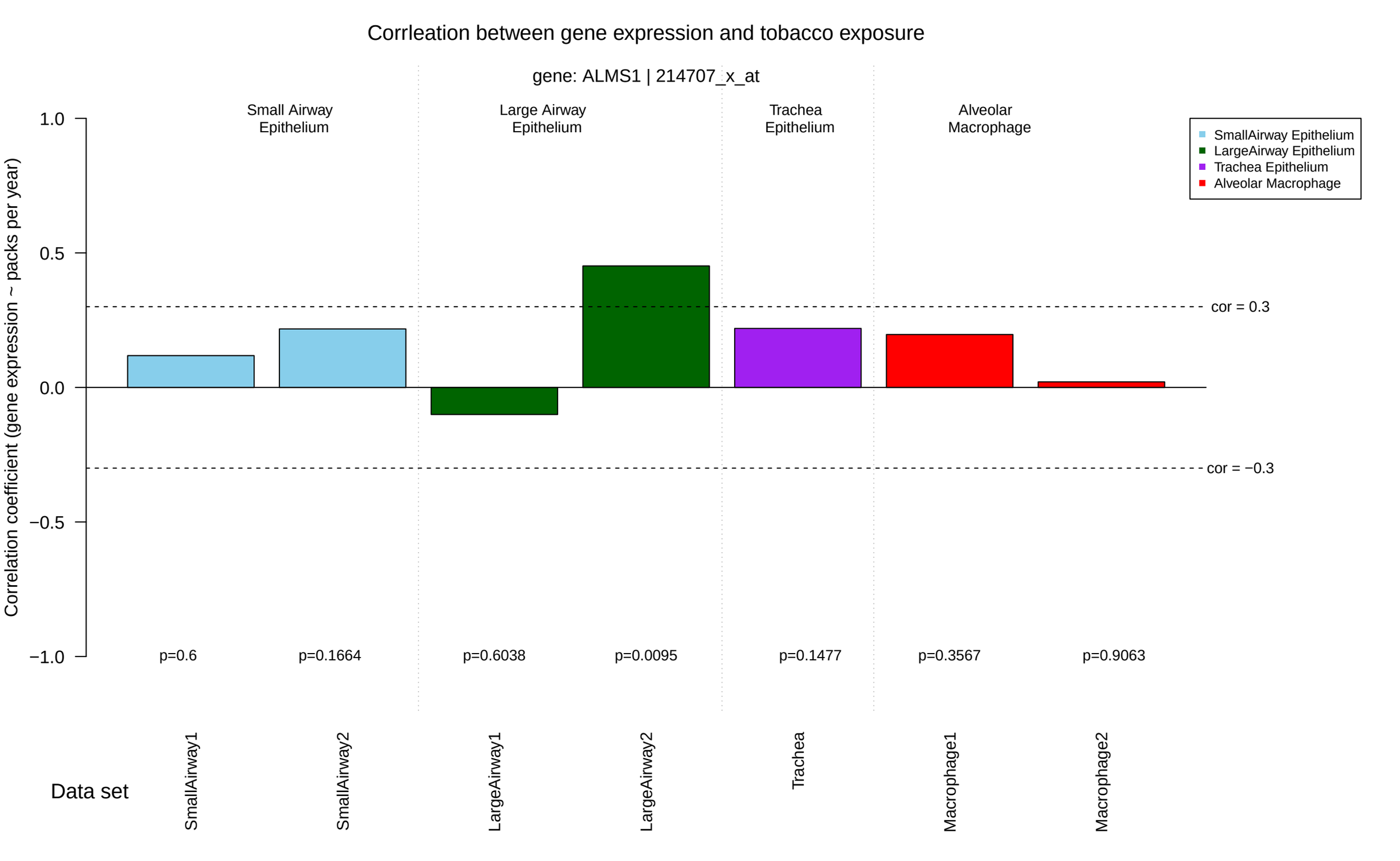

set: LargeAirway1 (p=0.8123)
ALMS1 | 214707_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.1767)
ALMS1 | 214707_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.082)
ALMS1 | 214707_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.1216)
ALMS1 | 214707_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.3576)
ALMS1 | 214707_x_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: ALMS1 | 214707_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.6
p=0.1664
p=0.6038
p=0.0095
p=0.1477
p=0.3567
p=0.9063
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2